

IMO(32nd) – 1991

First Day

1. Prove for each triangle ABC the inequality $\frac{1}{4} < \frac{IA \cdot IB \cdot IC}{l_A l_B l_C} \leq \frac{8}{27}$ where I is the incenter and l_A, l_B, l_C are the lengths of the angle bisector of ABC.
2. Let $n > 6$ and let $a_1 < a_2 < \dots < a_k$ be all natural numbers that are less than n and relatively prime to n . Show that if $a_1, a_2, \dots, a_k$ is an arithmetic progression, then n is a prime number or a natural power of two.
3. Let $S = \{1, 2, 3, \dots, 280\}$. Find the minimal natural number n such that in any n -elements subset of S there are five numbers, that are pair wise relatively prime.

Second Day

4. Suppose G is a connected graph with n edges. Prove that it is possible to label the edges of G from 1 to n in such a way that in every vertex v of G with two or more incident edges, the set of numbers labeling those edges has no common divisor greater than 1.
5. Let ABC be a triangle and M an interior point of ABC . Show that at least one of the angle $\angle MAB, \angle MBC, \angle MCA$ is less than or equal to 30° .
6. Given a real number $a > 1$, construct an infinite and bounded sequences $x_0, x_1, x_2, \dots$ such that for all natural numbers i and j , $i \neq j$, the following inequality holds: $|x_i - x_j| |i - j|^a \geq 1$.

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